3.4.5 Systems architecture

Content	Additional information
Explain the role and operation of main memory and the following major components of a central processing unit (CPU) within the Von Neumann architecture: • arithmetic logic unit • control unit • clock • register • bus.	A bus is a collection of wires through which data/signals are transmitted from one component to another. Knowledge of specific registers is not required.
Explain the effect of the following on the performance of the CPU: • clock speed • number of processor cores • cache size.	
Understand and explain the Fetch-Execute cycle.	The CPU continually reads instructions stored in main memory and executes them as required: • fetch: the next instruction is fetched to the CPU from main memory • decode: the instruction is decoded to work out what it is • execute: the instruction is executed (carried out). This may include reading/ writing from/to main memory.

Content	Additional information
Understand the different types of memory within a computer: • RAM • ROM • Cache • Register. Know what the different types of memory are used for and why they are required.	
Understand the differences between main memory and secondary storage. Understand the differences between RAM and ROM.	Students should be able to explain the terms volatile and non-volatile. Main memory will be considered to be any form of memory that is directly accessible by the CPU (except for cache and registers). Secondary storage is considered to be any non-volatile storage mechanism not directly accessible by the CPU.
Understand why secondary storage is required.	
Be aware of different types of secondary storage (solid state, optical and magnetic). Explain the operation of solid state, optical and magnetic storage. Discuss the advantages and disadvantages of solid state, optical and magnetic storage.	Students should be aware that SSDs use electrical circuits to persistently store data but will not need to know the precise details such as use of NAND gates.
Explain the term cloud storage.	Students should understand that cloud storage uses magnetic and/or solid state storage at a remote location.
Explain the advantages and disadvantages of cloud storage when compared to local storage.	
Understand the term embedded system and explain how an embedded system differs from a non-embedded system.	Students must be able to give examples of embedded and non-embedded systems.